MODULE INFORMATION SHEET

Department of Information Technology
Faculty of Humanities & Social Sciences – University of Ruhuna
2000 Level - Semester – II –2025-2026 Academic Year

I. Module Tittle and Code: ICT22513/ICT22614 Web Application Development

II. Module Coordinator: Dr. Kokila Ramanayaka (KR)

Lectures & lecture hours: Dr. Kokila Ramanayaka (KR) – 45 hours

Ms. S.A. Asra (SAA) – 15 hours

Module assistant/s:

III. Number of Credits Allocated: 03/04 Credits

IV. Volume of Learning: 200 Notional Hours [60 Direct Contact Hours + 140 Additional Learning/ Self Learning Hours]

V. Delivering Method: Lecturing, Practical, Discussion, Presentation, Tutorials, Group work

Direct contact learning hours: 60 hours: Lectures, videos followed by discussions, classroom discussions

Additional leaning hours: 140 hours

VI. Introduction to the Module:

The Web Application Development module is designed to provide students with a comprehensive understanding of building dynamic, data-driven web applications. Throughout this module, students will explore the core technologies involved in web development, including HTML, CSS, JavaScript, PHP, and MySQL. Emphasis is placed on developing both the front-end and back-end of web applications, and integrating databases. By the end of this module, students will have gained practical skills in designing, coding, and deploying fully functional web applications.

VII. Aims and objectives:

- Understand the principles of creating an effective web page, including an in-depth consideration of information architecture.
- Develop skills in analyzing the usability of a web site.
- Learn the language of the web: HTML and CSS.
- Develop basic programming skills using JavaScript and PHP

VIII. Intended Learning Outcomes (ILOs):

At the completion of this course, students will be able to:

- Explain the client-server architecture of the World Wide Web and its communication protocol HTTP/HTTPS
- Outline formats and languages used in modern web-pages
- Design web pages with HTML, CSS, JavaScript/DOM
- Design dynamic web-pages and web-applications using PHP and database handling
- Explain how to validate web pages according to W3C standard considering the need of clients and users
- Analyze and evaluate websites from the perspective on usability and availability

IX. STRUCTURE OF THE COURSE:
The module consists of lectures**,** practical, Discussion, Tutorials, group projects

X. Detailed Course content

	Content	**Learning activities to achieve intended learning outcomes**	**Suggested reference**	**Learning activities and time allocations**	**Direct conta ct hrs**
1	Introduction to the Internet and World Wide Web	Explain key concepts such as how the Internet works, the difference between the Internet and the WWW, protocols (HTTP, FTP), and the role of ISPs Explore different types of websites (educational, commercial, government) and performing specific searches using search operators to filter information Discuss the development of the Internet from ARPANET to modern-day broadband and the shift from Web 1.0 to Web 3.0	Harvey M. Deitel, Paul J. Deitel, Abbey Deitel, (2012). *"Internet and World Wide Web How to Program (International Version)"*, Pearson Education, ISBN-13: 978-0273764021.	Lectures	SAA (04 hrs)
2	Fundamentals of Hyper Text Markup Language (HTML)	Introduce the structure of an HTML document Write a basic HTML page using a text editor (e.g., Notepad++, VS Code) including headings, paragraphs, lists, and images in web page, learning how to properly structure and save the file	Ahmed, R. (2021). *Full Stack Web Development For Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP*. Independently published.	Lectures and Practical lab sessions	KR (06 hrs)

3	Introduction to Cascading Style Sheets	Explain the role of CSS in web development, the structure of CSS rules (selectors, properties, and values), and how CSS is linked to HTML documents Apply CSS in three different ways—inline, internal (using <style> tags in the <head>), and external (using a separate .css file) Explore different types of CSS selectors (e.g., type, class, ID, universal, group, descendant) and apply them in real-world examples	Ahmed, R. (2021). *Full Stack Web Development For Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP*. Independently published.	Lectures and Practical lab sessions	KR (03 hrs)
4	Client-side programming using JavaScript	Introduce JavaScript syntax and key concepts such as variables, loops, conditionals, and functions Create simple JavaScript programs such as displaying an alert, logging text to the console, and manipulating HTML content using JavaScript	Ahmed, R. (2021). *Full Stack Web Development For Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP*. Independently published.	Lectures and Practical lab sessions	KR (03 hrs)
5	Advanced CSS and Bootstrap	Introduce advanced CSS concepts, such as Flexbox for responsive layouts, Grid for multi-dimensional layouts, CSS transitions, and keyframe animations	Ahmed, R. (2021). *Full Stack Web Development For Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP*. Independently published.	Lectures and Practical lab sessions	SAA (08 hrs)
6	Server-side programming using PHP	Introduce the fundamentals of PHP, including syntax, variables, data types, control structures, and functions Install and configure a local server (e.g., XAMPP, MAMP, or WAMP) to run PHP on their machines Build a basic HTML form (e.g., login, registration, or contact form) and use PHP to process form data on the server side Explore session management using PHP sessions and cookies	Ahmed, R. (2021). *Full Stack Web Development For Beginners: Learn Ecommerce Web Development Using HTML5, CSS3, Bootstrap, JavaScript, MySQL, and PHP*. Independently published.	Lectures and Practical lab sessions	KR (08 hrs)

	Database Interactions	Explain the basics of PHP and how it interacts with databases, focusing on SQL, database connectivity (using MySQLi and PDO), and the principles of CRUD operations	Matthews, M. (2015). *PHP and MySQL Web Development: A Beginner's Guide (Beginner's Guide)* (1st ed.). McGraw Hill.	Lectures and Practical lab sessions	KR (08 hrs)
	Web Development Frameworks	Explain the latest frameworks used to develop web applications		Lectures and Practical lab sessions	KR (20 hrs)

XI. Assessments:

Two assignments (MCQ/Essay/Practical)

XII. General Policies and expectations:

Students should comply with the following **mandatory requirements** to be eligible for the summative assessment of the course.

i. 80% attendance in lectures
ii. Successful completion of all assessments
iii. Successful completion of all continuous and summative assessments